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Q3 a) P_n = number of mature salmon at the end of the n^{th} spawning cycle

Q_0 = produced in $n+1^{st}$ cycle

Assume :

$$Q_0 = \alpha P_n, \quad \text{for some constant } \alpha > 0$$

& rate of decline is $Q(t)$ is proportional both to $Q(t)$ and P_n.

$$\frac{d}{dt} Q(t) = -\beta Q(t) P_n \qquad \beta > 0$$

Hence

$$Q(t) = Q_0 e^{-\beta P_n t}$$

Over duration T surviving population is

$$Q(T) = Q_0 e^{-P_n/K} \quad \text{for}$$

$$K = \frac{1}{\beta T} > 0$$

All the spawning population die out so the population at the end of the $n+1^{st}$

$$P_{n+1} = \gamma Q(T) \quad \text{for } \gamma > 0$$

Defining $R_0 = \alpha\gamma$

$\therefore$ we find that

$$P_{n+1} = R_0 P_n e^{-P_n/K}$$

Q3 b) Define $R(P_n) = \dfrac{P_{n+1}}{P_n}$ & $R(P) = R_0 e^{-P/K}$

Clearly $R(P) < R_0$ i.e. R_0 is the max reproduction rate it occurs when $\beta = 0$ or for small $P \ll K$ where $e^{-P/K} \approx 1$

If $R_0 < 1$ then clearly $P_{n+1} < P_n$ and the salmon population will die out.

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Q2 c) cont. Fixing A in terms of initial value $p_0(0)$

$$p_0(t) = \frac{\beta}{\alpha+\beta}\left(1 - e^{-(\alpha+\beta)t}\right) + p_0(0)e^{(\alpha+\beta)t}$$

system is ergodic for α & β if not both are zero with

$$\pi_0 = \lim_{t\to\infty} p_0(t) = \frac{\beta}{\alpha+\beta}$$

$$\pi_1 = \lim_{t\to\infty} p_1(t) = \frac{\alpha}{\alpha+\beta}$$

Q2 d) 2min average waiting time → arrive $= \alpha$

3min average length of call → servicing $= \beta$

$$\alpha = \frac{1}{2}, \quad \beta = \frac{1}{3}$$

$$\rho = \frac{\alpha}{\beta} = \frac{1/2}{1/3} = \frac{3}{2}$$

$$\pi_1 = \frac{(3/2)^1}{1!} \frac{1}{1+3/2}$$

$$\pi_1 = \frac{3}{5}$$

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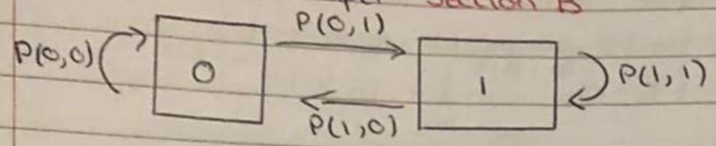
Q2 a)



$P(0,1) = \alpha \Delta t, \; P(0,0) = 1 - \alpha \Delta t, \; P(1,0) = \beta \Delta t, \; P(1,1) = 1 - \beta \Delta t$

Q2 b) $p_0(t+\Delta t) = p_0(t)(1-\alpha\Delta t) + p_1(t)\beta\Delta t$

$p_1(t+\Delta t) = p_0(t)(\alpha\Delta t) + p_1(t)(1-\beta\Delta t)$

$$\frac{dp_0}{dt} = \lim_{\Delta t \to 0} \frac{p_0(t+\Delta t) - p_0(t)}{\Delta t}$$

$$= -\alpha p_0(t) + \beta p_1(t)$$

$$\& \; \frac{dp_1}{dt} = \lim_{\Delta t \to 0} \frac{p_1(t+\Delta t) - p_1(t)}{\Delta t}$$

$$= \alpha p_0(t) - \beta p_1(t)$$

Q2 c) $\frac{d}{dt}(q_0 + q_1) = 0$

$$\frac{dp_0}{dt}(t) = -\alpha p_0(t) + \beta(1 - p_0(t))$$

$$= -(\alpha+\beta)p_0(t) + \beta$$

$$\frac{dp_0}{dt} + (\alpha+\beta)p_0 = \beta$$

$$\Rightarrow e^{(\alpha+\beta)t} + (\alpha+\beta)e^{(\alpha+\beta)t} p_0(t) = \frac{d}{dt}\left(e^{(\alpha-\beta)t} p_0(t)\right)$$

$$= pe^{(\alpha+\beta)t}$$

$$\frac{d}{dt}(p_0 + p_1) = 0$$

as expected $p_0 + p_1 = 1$

$$\frac{dp_0}{dt} + (\alpha+\beta)p_0 = \beta$$

Hence $\frac{d}{dt}\left(e^{(\alpha+\beta)t} p_0\right) = \beta e^{(\alpha+\beta)t}$

$$p_0(t) = \frac{\beta}{\alpha+\beta} + Ae^{-(\alpha+\beta)t}$$

some constant A

$\rightarrow$

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Q1 d) $\dfrac{dx_1}{dt} = x_2 + u$ $\qquad y = x_1 - x_2$

$y = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$\dfrac{dx_2}{dt} = bx_1 + x_2$

$$\frac{dx}{dt} = \begin{pmatrix} 0 & 1 \\ b & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u$$

i) $W_o = \begin{pmatrix} C \\ CA \end{pmatrix}$

$A = \begin{pmatrix} 0 & 1 \\ b & 1 \end{pmatrix} \qquad C = \begin{pmatrix} 1 & -1 \end{pmatrix}$

$$CA = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ b & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -b & 0 \end{pmatrix}$$

$$\begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -b & 0 \end{pmatrix}$$

The rows are linearly independent if $b \neq 0$

$\therefore$ the system is observable.

ii) $W_r = \begin{pmatrix} B & AB \end{pmatrix}$

$A = \begin{pmatrix} 0 & 1 \\ b & 1 \end{pmatrix} \qquad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$$AB = \begin{pmatrix} 0 & 1 \\ b & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ b \end{pmatrix}$$

$$W_r = \begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix}$$

$$\det(W_r) = \frac{1}{b}$$

$\therefore$ the system is reachable if $b \neq 0$

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Q1(c) cont.

$$S_{n+1} = AS_n + B$$

$$S_1 = AS_0 + B$$

$$S_2 = AS_1 + B$$

$$= A(AS_0 + B) + B$$

$$S_n = A^n S_0 + B(1 + A + \dots + A^{n-1})$$

$$S_n = A^n S_0 + B\left(\frac{1-A^n}{1-A}\right)$$

$$\Rightarrow P_n = \frac{1}{A^n S_0 + B_1 \frac{-A^n}{1-A}}$$

$$= \frac{1}{A^n P_0^{-1} + B\frac{1-A^n}{1-A}}$$

Hence we obtain the solution to Verhust solution

$$P_n = \frac{1}{\left(\frac{1-A^n}{1-A}\right)B + A^n P_0^{-1}}$$

Letting $r > 0$ so that $A < 1$

$$\lim_{n\to\infty} P_n \frac{1}{\frac{B}{1-A}} = \frac{1-A}{B}$$

$$= \frac{1 - \frac{1}{K+r}}{\frac{r}{K(1+r)}}$$

Thus the population has a liminting size of K independent of its initial size P_0.

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(Q1b iii) cont.

$$0.7\pi_2 = 0.7\pi_1$$
$$\pi_2 = \pi_1$$
$$0.1\pi_2 = 0.1\pi_3$$
$$\pi_2 = \pi_3$$
$$\pi_2(1+1+1) = 1$$
$$3\pi_2 = 1$$
$$\pi_2 = \frac{1}{3}$$

Equilibrium probabilities $= \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

Q1 c) $P_{n+1} = R(P_n)P_n$ ①

$$R(P) = \frac{1+r}{1+r\frac{P}{K}} \qquad r, K > 0$$

If $P_n = K$ then the ratio $R(K) = 1$ and $P_{n+1} = 1$

i.e. the model has a fixed point

$$P_{n+1} = P_n = P^*$$

$P^* = 0$ is a trivial fixed point $R(P^*) = 1$

$$\frac{1+r}{1+r^P/K} = 1 \Rightarrow P^* = K$$

If $0 < P \ll K$ $R(P) \simeq 1+r$ and population will grow since $r > 0$

To solve ① let $S_n = \frac{1}{P_n}$ and sub into ①

$$\frac{1}{S_{n+1}} = \frac{1+r}{1+{}^r/S_nK} \frac{1}{S_{n+1}} = \frac{1+r}{S_n + {}^r/K}$$

$$S_{n+1} = \frac{S_n}{1+r} + \frac{r}{K(1+r)}$$

$$S_{n+1} = \frac{1}{1+r} S_n + \frac{r}{K(1+r)}$$

$S_0 = \frac{1}{P_0}$ Let $A = \frac{1}{1+r}$, $B = \frac{r}{K(1+r)}$

→

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Q1a) i) $p_K = \frac{(\alpha t)^K}{K!} e^{-\alpha t}$ $K = 0,1,2,\ldots$

$t = 10$ $p_K = \frac{(10\alpha)^K}{K!} e^{-10\alpha}$ $K = 0,1,2,\ldots$

ii) Variance $\sigma^2 = \sum_{K \geq 0} (K-\mu)^2 p_K \geq 0$

$$= \sum_{K=0}^{\infty} \left(K - K\frac{(\alpha t)^K}{K!} e^{-\alpha t}\right)^2 \frac{(\alpha t)^K}{K!} e^{-\alpha t}$$

$t = 10$

$$= \sum_{K=0}^{\infty} \left(K - K\frac{(10\alpha)^K}{K!} e^{-10\alpha}\right)^2 \frac{(10\alpha)^K}{K!} e^{-10\alpha}$$

Standard deviation σ

$$= \sqrt{\sum_{K=0}^{\infty} \left(K - K\frac{(10\alpha)^K}{K!} e^{-10\alpha}\right)^2 \frac{(10\alpha)^K}{K!} e^{-10\alpha}}$$

Q1b) i) $P =$

$$\begin{array}{c c} & \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \end{array} & \begin{pmatrix} 0.3 & 0.7 & 0 \\ 0.7 & 0.2 & p \\ 0 & 0.1 & q \end{pmatrix} \end{array}$$

$0.7 + 0.2 + p = 1$

$p = 0.1$

$0.1 + q = 1$

$q = 0.9$

ii) Yes the process is ergodic as you can travel to each state using any of the states.

Q1b) iii) There is unique equilibirium probabilities

$$(\pi_1 \ \pi_2 \ \pi_3) \begin{pmatrix} 0.3 & 0.7 & 0 \\ 0.7 & 0.2 & 0.1 \\ 0 & 0.1 & 0.9 \end{pmatrix}$$

$0.3\pi_1 + 0.7\pi_2 = \pi_1$

$0.7\pi_1 + 0.2\pi_2 + 0.1\pi_3 = \pi_2$

$0.1\pi_2 + 0.9\pi_3 = \pi_3$

$\pi_1 + \pi_2 + \pi_3 = 1$

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Q4 a) $\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} -p_0 - p_1 & p_1 \\ p_2 & -p_2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} b_0 \\ 0 \end{pmatrix} u$

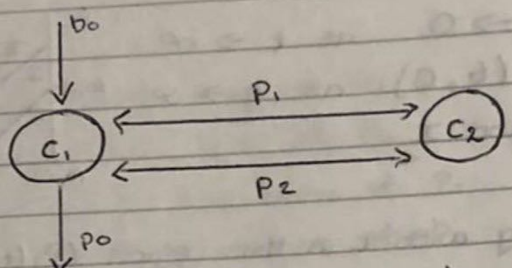
$y = c_1$

c_1, c_2 are concentrations of the drug in each compartment p_0, p_1, p_2

b_0 are rate constants

u is the flow rate of the drug into compartment 1

y is the concentration of the drug into compartment 2.



$\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} (-p_0 - p_1)c_1 + p_1 c_2 \\ p_2 c_1 - p_2 c_2 \end{pmatrix} + \begin{pmatrix} b_0 u \\ 0 \end{pmatrix}$

$= \begin{pmatrix} -p_0 c_1 - p_1 c_1 + p_1 c_2 + b_0 u \\ p_2 c_1 - p_2 c_2 \end{pmatrix}$

$\dot{c}_1 = -p_0 c_1 - p_1 c_1 + p_1 c_2 + b_0 u$

$\dot{c}_2 = p_2 c_1 - p_2 c_2$

Q4 b) $\frac{dx}{dt} = Ax + Bu, \; y = Cx + Du$

$A = \begin{pmatrix} -p_0 - p_1 & p_1 \\ p_2 & -p_2 \end{pmatrix}, \; B = \begin{pmatrix} b_0 \\ 0 \end{pmatrix}, \; C = (1, 0)$

$W_0 = \begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -p_0 - p_1 & p_1 \end{pmatrix}$

The rows are linearly independent if $p_1 \neq 0$.

Q4 c) $u = -Kc + K_r r, \quad K = (K_1, K_2), \; c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

K_1, K_2, K_r = control parameters

$r = 2$ = desired output

eigenvalues $\rightarrow \lambda_1 = -1, \; \lambda_2 = -1$

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Q3b) cont. Sub $P_1 = 6 - P_2$ into $\dot{P}_2 = 0$

$$P_2(2 - 6 + P_2 - P_2) = 0$$

$$P_2(-4) = 0$$

$P_2 = 0$ $\quad 0(-4) = 0$

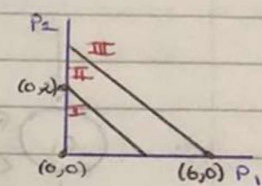
$$0 = 0$$

$P_1 = 6 - 0 = 6$

Equilibrium point $= (6, 0)$

Q3c) $P_1(t) \to 6$, $P_2(t) \to 0$ as $t \to \infty$

$(P_1(t), P_2(t)) \to (6, 0)$ as $t \to \infty$



Consider (P_1, P_2) in region I

$\dot{P}_1 > 0$, $\dot{P}_2 > 0$

P_1 and P_2 are increasing with t in this region. $(P_1(t), P_2(t))$ must cross into region II on the line $P_1 + P_2 = 2$ where $\dot{P}_2 = 0$ & $\dot{P}_1 > 0$

so $(0,0)$ and $(0,2)$ are unstable fixed points.

Consider (P_1, P_2) in region II

$\dot{P}_1 > 0$, $\dot{P}_2 < 0$

Hence P_1 grows and is bounded above by 6 & P_2 decreases and is bounded below by 0.

So the curve is trapped between the lines $P_1 + P_2 = 2$ and $P_1 + P_2 = 6$.

If $P_1(t^*) + P_2(t^*) = 2$ at some time t^* then $\dot{P}_1(t^*) = 0$

But $\ddot{P}_1(t^*) = 4P_1(t^*)\dot{P}_2(t^*) > 0$

so that $t = t^*$ is a local minimum of $P_1(t)$ which is impossible as P_1 is increasing in region II

Similarly if $P_1(t^*) + P_2(t^*) = 6$ then $\dot{P}_2(t^*) = 0$

and $\ddot{P}_2(t^*) = P_2(t^*)\dot{P}_1(t^*) < 0$

So that $t = t^*$ is a local maximum of $P_2(t)$ which is impossible since P_2 is decreasing within region II

Hence $P_1(t) \to 6$ and $P_2(t) \to 0$ with the curve remaining with region II at all times.

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Q1 d) ii) A linear system is reachable if for any $x_0, x_F \in \mathbb{R}^n$ there exists a time $T > 0$ and $u: [0,T] \to \mathbb{R}$ such that the corresponding solution satisfies $x(0) = x_0$ and $x(T) = x_F$.

Section B

Q3 a)
$$\frac{dP_1}{dt} = 4P_1(6 - P_1 - P_2)$$
$$\frac{dP_2}{dt} = P_2(2 - P_1 - P_2)$$

P_1 = population size of P_1

P_2 = population size of P_2

6 = maximum population of P_1

2 = maximum population of P_2

$4P_1$ = biotic population for P_1

$1P_2$ = biotic population for P_2

Q3 b) $\dot{P}_1 = 0$, $\dot{P}_2 = 0$ -> Equilibrium parts

$\dot{P}_1 = 0 \Rightarrow 4P_1(6 - P_1 - P_2) = 0$

$P_1 = 0$ $\quad 4(0)(6 - 0 - P_2) = 0$

$0 = 0$

$P_1 = 6 - P_2$ $\quad 4(6 - P_2)(6 - 6 + P_2 - P_2) = 0$

$4(6 - P_2)(0) = 0$

$0 = 0$

Sub $P_1 = 0$ into $\dot{P}_2 = 0$

$P_2(2 - P_2) = 0$

$P_2 = 0$ $\quad 0(2 - 0) = 0$

$0 = 0$

Equilibrium point = $(0,0)$

$P_2 = 2$ $\quad 2(2 - 2) = 0$

$2(0) = 0$

$0 = 0$

Equilibrium point = $(0,2)$

$\Leftarrow$

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Q1c) ii) Equilibrium points:

$$\frac{dx}{dt} = 0 \quad \& \quad \frac{dy}{dt} = 0$$

$$\frac{dx}{dt} = 0 \Rightarrow x = 0$$

$$-by + a = 0$$

$$y = \frac{a}{b}$$

sub $x=0$ into $\frac{dy}{dt} = 0$

$$-cy = 0 \Rightarrow y = 0$$

sub $y = \frac{a}{b}$ into $\frac{dy}{dt} = 0$

$$\frac{a}{b}(bx - c) = 0 \Rightarrow ax - \frac{ac}{b} = 0$$

$$\Rightarrow x = \frac{c}{b}$$

The equilibrium points are:

$$(x, y) = (0, 0), \left(\frac{c}{b}, \frac{a}{b}\right)$$

Q1d) i)

$$\frac{dx_1}{dt} = a_1 x_1 + a_2 x_2 + b_1 u$$

$$\frac{dx_2}{dt} = a_3 x_1 + a_4 x_2 + b_2 u$$

$$y = c_1 x_1 + c_2 x_2$$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ y \end{pmatrix} = \begin{pmatrix} a_1 x_1 + a_2 x_2 + b_1 u \\ a_3 x_1 + a_4 x_2 + b_2 u \\ c_1 x_1 + c_2 x_2 \end{pmatrix}$$

$$= \begin{pmatrix} a_1 & a_2 & b_1 \\ a_3 & a_4 & b_2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ u \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ c_1 & c_2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

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Q1 b) ii) cont.

$0.7\pi_3 + 0.2\pi_2 = \pi_3$

$\pi_2 = 1.5\pi_3$

$\pi_3(1 + 1.5 + 1) = 1$

$3.5\pi_3 = 1$

$\pi_3 = \frac{2}{7} \Rightarrow \pi_1 = \frac{2}{7}$

$0.3\pi_1 = 0.2\pi_2$

$\pi_1 = \frac{2}{3}\pi_2$

$0.7\pi_3 + 0.2\pi_2 = 0.3\pi_3$

$\pi_3 = \frac{2}{3}\pi_2$

$\pi_2\left(\frac{2}{3} + 1 + \frac{2}{3}\right) = 1$

$\frac{7}{3}\pi_2 = 1$

$\pi_2 = \frac{3}{7}$

Equilibrium probabilities $= \left(\frac{2}{7}, \frac{3}{7}, \frac{2}{7}\right)$.

Q1 c) i)

$x(t)$ = prey population

$y(t)$ = predator population

$\frac{dx}{dt} = x(-by + a)$

$\frac{dy}{dt} = y(bx - c)$

a = prey growth rate without predators

c = predator death rate without prey

by = models predator suppression of prey's growth

bx = models prey enhancement of predators growth.

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Section A

Q1a) A finite ergodic Markov process with transition probabilty matrix P is defined by:

$$\pi_i \equiv \lim_{m \to 0} P_m(j,i)$$

where π_i are equilibrium probabilities

$$\pi_i = (\pi_0, \pi_1, \pi_2, \pi_3, \pi_4)$$

$$\pi = \pi P$$

$$\pi_i = \pi_0 P(0,i) + \pi_1 P(1,i) + \pi_2 P(2,i) + \pi_3 P(3,i) + \pi_4 P(4,i)$$

$$(\pi_0, \pi_1, \pi_2, \pi_3, \pi_4) = (\pi_0, \pi_1, \pi_2, \pi_3, \pi_4) P$$

The rows of P^m become similar as m increases. This shows that $P_m(i,j)$ = prob($i \to j$, m steps) which leads to an Ergodic system.

Q1b) i)

$$P = \begin{array}{c|c} & \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \end{array} & \begin{pmatrix} 0 & p & 0.7 \\ 0 & 0.8 & q \\ 1 & 0 & 0 \end{pmatrix} \end{array} = \begin{pmatrix} 0 & 0.3 & 0.7 \\ 0 & 0.8 & 0.2 \\ 1 & 0 & 0 \end{pmatrix}$$

$$p + 0.7 = 1$$
$$p = 0.3$$
$$q + 0.8 = 1$$
$$q = 0.2$$

Q1b) ii) Yes the process is ergodic as you can travel to each states using the other states i.e. there is some connection between the states where we can visit each state. . . — . . .

Q1b) iii) There is unique equilibrium probabilities.

$$(\pi_1, \pi_2, \pi_3) \begin{pmatrix} 0 & 0.3 & 0.7 \\ 0 & 0.8 & 0.2 \\ 1 & 0 & 0 \end{pmatrix}$$

$$\pi_3 = \pi_1$$

$0.3\pi_1 + 0.8\pi_2 = \pi_2$ $\qquad$ $0.3\pi_2 + 0.7\pi_3 = \pi_1$

$0.7\pi_1 + 0.2\pi_2 = \pi_3$ $\qquad$ $0.8\pi_2 + 0.2\pi_3 = \pi_2$

$\pi_1 + \pi_2 + \pi_3 = 1$ $\qquad$ $\pi_1 = \pi_3$

$\Leftarrow$

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...4 b) cont.

$$K_r = \frac{1}{-2k_1 - 4k_2 - 1} \begin{pmatrix} 2 & 1-2k_1 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

$$= \frac{1}{-2k_1 - 4k_2 - 1} (4)$$

$$= \frac{4}{-2k_1 - 4k_2 - 1}$$

$$u = -k_1 c_1 - k_2 c_2 - 2K_r$$

$$= -k_1 c_1 - k_2 c_2 - 2\left(\frac{4}{-2k_1 - 4k_2 - 1}\right)$$

where $k_1 = 3$, $k_2 = -\frac{11}{4}$

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Q4 b) cont.

$u = -Kc + K_r r$

$= -K_1 c_1 - K_2 c_2 + 2K_r$

Closed loop system becomes

$$\frac{dc}{dt} = (A - BK)c + BK_r r$$

$$= \begin{pmatrix} 1-2K_1 & -1-2K_2 \\ -2 & 1 \end{pmatrix} c + 2\begin{pmatrix} 2K_r \\ 0 \end{pmatrix}$$

$y = Cc = c_2$

$$\det(A - BK - \lambda I) = \begin{vmatrix} 1-2K_1-\lambda & -1-2K_2 \\ -2 & 1-\lambda \end{vmatrix}$$

$$= (1-2K_1-\lambda)(1-\lambda) - (-1-2K_2)(-2)$$

$$= 1 - \lambda - 2K_1 + 2K_1\lambda - \lambda + \lambda^2 - (2 + 4K_2)$$

$$= \lambda^2 - 2\lambda + 2K_1\lambda + 1 - 2K_1 - 2 - 4K_2$$

$$= \lambda^2 + (-2 + 2K_1)\lambda + 1 - 2K_1 - 2 - 4K_2$$

Eigenvalues: $\lambda_1 = -2$, $\lambda_2 = -2$

$$(\lambda+2)(\lambda+2) = \lambda^2 + 4\lambda + 4$$

$$\lambda^2 + (-2+2K_1)\lambda + 1 - 2K_1 - 2 - 4K_2 = \lambda^2 + 4\lambda + 4$$

$-2 + 2K_1 = 4$ $\qquad$ $1 - 2K_1 - 2 - 4K_2 = 4$

$2K_1 = 6$ $\qquad$ $1 - 2(3) - 2 - 4K_2 = 4$

$K_1 = 3$ $\qquad$ $-4K_2 = 11$

$$K_2 = -\frac{11}{4}$$

Control parameter K_r:

$$C(A-BK)^{-1}B = (0 \;\; 1)\begin{pmatrix} 1-2K_1 & -1-2K_2 \\ -2 & 1 \end{pmatrix}^{-1}\begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

$$\frac{1}{(1-2K_1)(1) - (-1-2K_2)(-2)}\begin{pmatrix} 1 & 1+2K_2 \\ 2 & 1-2K_1 \end{pmatrix}$$

$$\frac{1}{-2K_1 - 4K_2 - 1}\begin{pmatrix} 1 & 1+2K_2 \\ 2 & 1-2K_1 \end{pmatrix}$$

$$K_r = \frac{1}{-2K_1 - 4K_2 - 1}\begin{pmatrix} 1 & 1+2K_2 \\ 2 & 1-2K_1 \end{pmatrix}(0 \;\; 1)\begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

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Q4 a) $A = \begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix}$ $B = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ $C = (0 \quad 1)$

Observable

$W_o = \begin{pmatrix} C \\ CA \end{pmatrix}$

$CA = (0 \quad 1)\begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix}$

$= (-2 \quad 1)$

$W_o = \begin{pmatrix} 0 & 1 \\ -2 & 1 \end{pmatrix}$

The rows are linearly independent

$\therefore$ the system is obserable

Reachable

$W_r = (B \quad AB)$

$AB = \begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

$= \begin{pmatrix} 2 \\ -4 \end{pmatrix}$

$W_r = \begin{pmatrix} 2 & 2 \\ 0 & -4 \end{pmatrix}$

$\det(W_r) = -\frac{1}{8}$

$\therefore$ the system is reachable.

Q4 b) $\frac{dc}{dt} = \begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \end{pmatrix} u$, $y = (0 \quad 1) c_2$

$A = \begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, $C = (0 \quad 1)$

$W_r = (B \quad AB) = \begin{pmatrix} 2 & 2 \\ 0 & -4 \end{pmatrix}$

$W_o = \begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -2 & 1 \end{pmatrix}$

$\rightarrow$

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Q3 c) If $R_0 > 1$ then an equilibrium fixed point occurs i.e. $R(p^*) =$ for $p^* = K \ln R_0 > 0$

which converges over time to the fixed point for

$1 < R_0 < e^2$ i.e. fixed point is stable.

Q3 d) Consider $F(P) = R_0 P e^{-P/K} = PR(P)$

for $P = p^* + \varepsilon$ for small ε

from Taylors theorem

$$F(p^* + \varepsilon) = F(p^*) + F'(p^*)\varepsilon + O(\varepsilon^2)$$
$$= p^* + F'(p^*)\varepsilon + O(\varepsilon^2)$$

Let $P_0 = p^* + \varepsilon$ then

$$P_1 = F(p^* + \varepsilon) \simeq p^* + F'(p^*)\varepsilon$$

so that $|P_1 - p^*| \simeq |F'(p^*)||P_0 - p^*|$ similarly over n cycles we have $|P_n - p^*| \simeq |F'(p^*)|^n |P_0 - p^*|$

Hence

$$|F'(p^*)| < 1$$

the population converges over time to a fixed point p^*

using $F'(P) = R_0(1 - P/K)e^{-P/K}$

$$F'(p^*) = 1 - \frac{p^*}{K}$$

For $1 < R_0 < e$ we have $p^* < K$ with $0 < F'(p^*) < 1$.

For $R_0 > e$ we have $p^* > K$ so that $F'(p^*) < 0$

Convergence occurs provided $F'(p^*) > -1$ so that

$p^* > 2K$ i.e. $R_0 < e^2 = 7.389$

For $R_0 > e^2$ period doubling effects occur and for large R_0 chaotic behaviour occurs.

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Q1d ii) cont $W_r = (B \quad AB)$

$= \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$

$\det(W_r) = -1$

$\therefore$ the system is reachable

Section B

Q2a) i) max queue size $= r$

$\alpha_r = \beta_0 = 0$

The probability of arrival and servicing is independent of the queue size. Only one customer can arrive or be serviced in one time interval.

Hence $\alpha_r = \beta_0 = 0$

Q2a) ii) Show in two steps:

(1) $\pi_K = \rho \pi_{K-1} \Rightarrow \pi_K = \rho^2 \pi_{K-2} = \dots = \rho^K \pi_0$

(2) $\pi_0 = \dfrac{1-\rho}{1-\rho^{r+1}}$, $\rho \neq 1$, $\pi_0 = \dfrac{1}{1+r}$, $\rho = 1$

(1) by induction:

Using the transition matrix P we have:

$\pi = \pi P$, $\pi = (\pi_0, \dots, \pi_r)$

$\pi_0 = \pi_0(1-p_1) + \pi_1 p_2$

$0 = -\pi_0 p_1 + \pi_1 p_2$

$\pi_1 p_2 = \pi_0 p_1$

$\pi_1 = \dfrac{p_1}{p_2} \pi_0 = \rho \pi_0$

(1) holds for $K = 1$

(2) Assume $\pi_{K-1} = \rho \pi_{K-2}$ holds then we have $(K-1)^{st}$ element of $\pi = \pi P$ gives

$\pi_{K-1} = \pi_{K-2} p_1 + \pi_{K+1}(1-p_1-p_2) + \pi_K p_2$

$\Rightarrow p_2 \pi_K = -\pi_{K-2} p_1 + (p_1+p_2)\pi_{K-1}$

$\Rightarrow \{ \pi_{K-1} = \rho \pi_{K-2} \Rightarrow p_2 \pi_{K-1} = p_1 \pi_{K-2} \}$

$p_2 \pi_K = \dfrac{-p_2 \pi}{K-1} + (p_1+p_2)\pi_{K-1}$

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Q1 c) $\frac{dS}{dt} = -\alpha S I$, $S(0) = S_0$, $\alpha > 0$

$\frac{dR}{dt} = \beta I$, $R(0) = 0$, $\beta > 0$ $\quad \rho = \frac{\beta}{\alpha}$

$$\frac{dS}{dR} = \frac{\dot{S}}{\dot{R}} = \frac{-\alpha S I}{\beta I} = -\frac{S}{\rho}$$

Hence

$S(R) = S_0 e^{-R/\rho}$ & $R(S) = -\rho \ln\left(\frac{S}{S_0}\right)$

Q1 d) $\frac{dx_1}{dt} = x_2$, $\frac{dx_2}{dt} = bx_1 + x_2 + u$, $y = x_1 + x_2$

i) $$\frac{dx}{dt} = \begin{pmatrix} x_2 \\ bx_1 + x_2 + u \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 \\ b & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$$

$$y = (1 \ \ 1)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$A = \begin{pmatrix} 0 & 1 \\ b & 1 \end{pmatrix}$ $\quad C = (1 \ \ 1)$

$$CA = (1 \ \ 1)\begin{pmatrix} 0 & 1 \\ b & 1 \end{pmatrix}$$

$$= (b \ \ 2)$$

$$W_0 = \begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ b & 2 \end{pmatrix}$$

The rows are linearly indepedent if $b \neq 2$

∴ the system is observable

ii) $A = \begin{pmatrix} 0 & 1 \\ b & 1 \end{pmatrix}$ $\quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$AB = \begin{pmatrix} 0 & 1 \\ b & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Q1 b) i cont

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Q1b)iii) cont.

$$0.3\pi_1 + 0.2\pi_2 + 0.5\pi_3 = \pi_3$$

$$0.3\pi_1 + 0.2\pi_2 = 0.5\pi_3$$

$$0.3(^{25}/_{19}) + 0.2\pi_2 = 0.5\pi_3$$

$$0.2\pi_2 = \frac{2}{19}\pi_3$$

$$\pi_2 = \frac{10}{19}\pi_3$$

$$0.5\pi_1 + 0.3\pi_2 + 0.5\pi_3 = \pi_1$$

$$-0.5\pi_1 + 0.3\pi_2 = -0.5\pi_3$$

$$-0.5\pi_1 + 0.12\pi_1 = -0.5\pi_3$$

$$0.38\pi_1 = 0.5\pi_3$$

$$\pi_1 = \frac{25}{19}\pi_3$$

$$\pi_3\left(\frac{25}{19} + \frac{10}{19} + 1\right) = 1$$

$$\frac{54}{19}\pi_3 = 1$$

$$\pi_3 = \frac{19}{54}$$

Equilibrium probabilities $= \left(\frac{25}{54}, \frac{5}{27}, \frac{19}{54}\right)$

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Q1b) i) $P = \begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & 0.5 & 0.2 & 0.3 \\ 1 & 0.3 & 0.5 & q \\ 2 & 0.5 & 0 & p \end{array}$

$q = 1 - 0.5 - 0.3 = 0.2$

$p = 1 - 0.5 = 0.5$

Q1b) ii) Yes the system is ergodic as there is a path you can travel to get to each node.

Q1b) iii) $(\pi_1 \; \pi_2 \; \pi_3)\begin{pmatrix} 0.5 & 0.2 & 0.3 \\ 0.3 & 0.5 & 0.2 \\ 0.5 & 0 & 0.5 \end{pmatrix}$

$0.5\pi_1 + 0.3\pi_2 + 0.5\pi_3 = \pi_1$

$0.2\pi_1 + 0.5\pi_2 + 0\pi_3 = \pi_2$

$0.3\pi_1 + 0.2\pi_2 + 0.5\pi_3 = \pi_3$

$\pi_1 + \pi_2 + \pi_3 = 1$

$0.2\pi_1 = 0.5\pi_2$

$\pi_1 = 2.5\pi_2$

$0.5(2.5\pi_2) + 0.3\pi_2 + 0.5\pi_3 = 2.5\pi_2$

$0.5\pi_3 = 0.95\pi_2$

$\pi_3 = 1.9\pi_2$

$\pi_2(2.5 + 1 + 1.9) = 1$

$\pi_2 = \frac{1}{5.4} = \frac{5}{27}$

$0.2\pi_1 + 0.5\pi_2 = \pi_2$

$0.5\pi_2 = 0.2\pi_1$

$\pi_2 = 0.4\pi_1$

$0.3\pi_1 + 0.2(0.4\pi_1) + 0.5\pi_3 = \pi_3$

$0.38\pi_1 = 0.5\pi_3$

$\pi_3 = 0.76\pi_1$

$\pi_1(1 + 0.4 + 0.76) = 1$

$\pi_1 = \frac{1}{2.16} = \frac{25}{54}$

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Q 1 a) $f(t) = \begin{cases} 1/b-a & , a \leq t \leq b, \\ 0 & , t < a, t, > b \end{cases}$

(i) P_K = Prob (state K occurs)

K = time = t

$$\sum_{a \leq t \leq b}^{\infty} \frac{1}{b-a}$$

(ii) Variance & Standard deviation

σ^2 ... σ

$$\sigma^2 = \langle t^2 \rangle - \mu^2$$

$$\mu = \langle t \rangle = \sum_{t \geq 0} t \, p_t$$

$$= \sum_{a \leq t \leq b}^{\infty} \frac{t}{b-a}$$

$$\langle t^2 \rangle = \sum_{a \leq t \leq b}^{\infty} \frac{t^2}{b-a}$$

$$\sigma^2 = \sum_{a \leq t \leq b}^{\infty} \frac{t^2}{b-a} - \sum_{a \leq t \leq b}^{\infty} \frac{t^2}{(b-a)^2}$$

$$\sigma^2 = t^2 \left(\frac{1}{b-a} - \frac{1}{(b-a)^2} \right) = \text{Variance}$$

$$\sigma = \sqrt{t^2 \left(\frac{1}{b-a} - \frac{1}{(b-a)^2} \right)} = \text{Standard deviation}$$

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Q2 b) cont.

Then $\pi_K = \frac{\rho^K}{K!} \cdot \frac{1}{1 + \rho + \frac{\rho^2}{2!} + \dots + \frac{\rho^r}{r!}}$

Q2 c)

phone exchange with 4 operators

24 people call exchange per hour

each phone call serviced in 5 minutes

$\alpha = \frac{24}{60} \text{ min}^{-1} = \frac{2}{5} \text{ min}^{-1}$

$\beta = \frac{1}{5} \text{ min}^{-1}$

$\rho = \frac{24/60}{1/5} = 2$

$r = 4$

$\pi_4 = \frac{2^4}{4!} \cdot \frac{1}{1 + 2 + \frac{2^2}{2!} + \frac{2^3}{3!} + \frac{2^4}{4!}}$

$\pi_4 = \frac{2}{3}\left(\frac{1}{7}\right)$

$\pi_4 = \frac{2}{21}$

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Q2a) ii) so (1) holds for K

$$\Rightarrow \pi_K = \rho^K \pi_0$$

We know $\sum_{K=0}^{r} \pi_K = 1$

$$\Rightarrow \sum_{K=0}^{r} \rho^K \pi_0 = 1$$

$$\Rightarrow \pi_0 = \frac{1}{\sum_{K=0}^{r} \rho^K}$$

$$\Rightarrow \pi_0 = \frac{1}{\frac{1-\rho^{r+1}}{1-\rho}} = \frac{1-\rho}{1-\rho^{r+1}}, \quad \rho \neq 1$$

$$\pi_0 = \frac{1}{r+1}, \quad \rho = 1$$

Q2 b) $\alpha_K = \alpha, \quad \beta_K = K\beta, \quad \alpha > 0, \quad \beta > 0$

Show π_K, $K = 0, 1, 2, \dots, r$ are given by

$$\pi_K = \frac{\rho^K}{K!} \frac{1}{1 + \rho + \frac{\rho^2}{2!} + \dots + \frac{\rho^r}{r!}} \qquad \text{where } \rho = \frac{\alpha}{\beta}$$

$$\pi_K = \frac{\rho^K}{K!} \pi_0$$

$$\pi_K = \frac{\rho^K}{K!} e^{-\rho}$$

Since $\sum_{K=0}^{r} \pi_K = 1$ we get $\pi_0 \sum_{K=0}^{r} \frac{\rho^K}{K!} = 1$

$$\pi_0 = \frac{1}{\sum_{K=0}^{r} \frac{\rho^K}{K!}}$$

$$\Rightarrow \pi_K = \frac{\rho^K}{K!} e^{-\rho} \frac{1}{\sum_{i=0}^{r} \frac{\rho^K}{i!}} \qquad K = 0, 1, \dots, r$$

$\leftarrow$

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Q2 a) α_K = poisson arrival pattern

β_K = poisson servicing pattern

K = queue size

Q2 a) i) max queue size = r

$\alpha_r = \beta_0 = 0$

The probability of arrival and servicing is independent of the queue size. Only one customer can arrive or be serviced in one time interval.

Hence $\alpha_r = \beta_0 = 0$

Q2 a) ii) Show in two steps:

(1) $\pi_K = \rho\pi_{K-1} \Rightarrow \pi_K = \rho^2\pi_{K-2} = \dots = \rho^K\pi_0$

(2) $\pi_0 = \dfrac{1-\rho}{1-\rho^{r+1}}$, $\rho \neq 1$, $\pi_0 = \dfrac{1}{1+r}$, $\rho = 1$

(1) by induction:

Using the transition matrix P we have

$\pi = \pi P$, $\pi = (\pi_0, \dots, \pi_r)$

$\pi_0 = \pi_0(1-p_1) + \pi_1 p_2$

$0 = -\pi_0 p_1 + \pi_1 p_2$

$\pi_1 p_2 = \pi_0 p_1$

$\pi_1 = \dfrac{p_1}{p_2}\pi_0 = \rho\pi_0$

(1) holds for K = 1

(2) Assume $\pi_{K-1} = \rho\pi_{K-2}$ holds then we have $(K-1)^{st}$ element of $\pi = \pi P$ gives

$\pi_{K-1} = \pi_{K-2}p_1 + \pi_{K+1}(1-p_1-p_2) + \pi_K p_2$

$\Rightarrow p_2\pi_K = -\pi_{K-2}p_1 + (p_1+p_2)\pi_{K-1}$

$\Rightarrow \{\pi_{K-1} = \rho\pi_{K-2} \Rightarrow p_2\pi_{K-1} = p_1\pi_{K-2}\}$

$p_2\pi_K = \dfrac{-p_2\pi}{K-1} + (p_1+p_2)\pi_{K-1}$

$= \pi_K = \dfrac{p_1}{p_2}\pi_{K-1} = \rho\pi_{K-1}$

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Q4 c) Control parameter K_r:

$$C(A-BK)^{-1}B = (1 \;\; 0)\begin{pmatrix} -p_0-p_1-b_0K_1 & p_1-b_0K_2 \\ p_2 & -p_2 \end{pmatrix}^{-1}\begin{pmatrix} b_0 \\ 0 \end{pmatrix}$$

$$\frac{1}{(-p_0-p_1-b_0K_1)(-p_2)-(p_1-b_0K_2)(p_2)}\begin{pmatrix} -p_2 & -p_1+b_0K_2 \\ -p_2 & -p_0-p_1-b_0K_1 \end{pmatrix}$$

$$\frac{1}{p_0p_2+p_1p_2+b_0K_1p_2-p_1p_2+b_0K_2p_2}\begin{pmatrix} -p_2 & -p_1+b_0K_2 \\ -p_2 & -p_0-p_1-b_0K_1 \end{pmatrix}$$

$$\frac{1}{p_2(p_0+b_0K_2+b_0K_1)}\begin{pmatrix} -p_2 & -p_1+b_0K_2 \\ -p_2 & -p_0-p_1-b_0K_1 \end{pmatrix}$$

$$C(A-BK)^{-1}B = (1 \;\; 0)\begin{pmatrix} -p_2 & -p_1+b_0K_2 \\ -p_2 & -p_0-p_1-b_0K_1 \end{pmatrix}\begin{pmatrix} b_0 \\ 0 \end{pmatrix}\frac{1}{p_2(p_0+b_0K_2+b_0K_1)}$$

$$= (-p_2 \quad -p_1b_0K_2)\begin{pmatrix} b_0 \\ 0 \end{pmatrix}\frac{1}{p_2(p_0+b_0K_2+b_0K_1)}$$

$$= (-p_2b_0)\frac{1}{p_2(p_0+b_0K_2+b_0K_1)}$$

$$C(A-BK)^{-1}B = \frac{-b_0}{p_0+b_0K_2+b_0K_1}$$

$$K_r = \frac{p_0}{b_0} + K_2 + K_1$$

$$u = -K_1c_1 - K_2c_2 + 2K_r$$
$$= -K_1c_1 - K_2c_2 + 2\left(\frac{p_0}{b_0} + K_2 + K_1\right)$$
$$= -K_1c_1 - K_2c_2 + \frac{2p_0}{b_0} + 2K_2 + 2K_1$$

$$u = K_1(2-c_1) + K_2(2-c_2) + \frac{2p_0}{b_0}$$

with $K_1 = \dfrac{2-p_1-p_2-p_0}{b_0}$, $K_2 = \dfrac{1}{b_0}\left(\dfrac{1}{p_2} - 2 + p_2 + p_1\right)$

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Q4 c) $\frac{dc}{dt} = \begin{pmatrix} -p_0 - p_1 & p_1 \\ p_2 & -p_2 \end{pmatrix} c + \begin{pmatrix} b_0 \\ 0 \end{pmatrix} u$, $y = (1 \;\; 0)\, c$

$A = \begin{pmatrix} -p_0 - p_1 & p_1 \\ p_2 & -p_2 \end{pmatrix}$, $B = \begin{pmatrix} b_0 \\ 0 \end{pmatrix}$, $C = (1, 0)$

$W_r = (B \;\; AB) = \begin{pmatrix} b_0 & -b_0(p_0 + p_1) \\ 0 & b_0 p_2 \end{pmatrix}$

$W_o = \begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -p_0 - p_1 & p_1 \end{pmatrix}$

$u = -Kc + K_r r$

$= -K_1 c_1 - K_2 c_2 + 2K_r$

Closed loop system becomes

$\frac{dc}{dt} = (A - BK)c + BK_r r$

$= \begin{pmatrix} -p_0 - p_1 - b_0 K_1 & p_1 - b_0 K_2 \\ p_2 & -p_2 \end{pmatrix} c + 2 \begin{pmatrix} b_0 K_r \\ 0 \end{pmatrix}$

$y = Cc$

$= (1 \;\; 0) c = c_1$

$\det(A - BK - \lambda I) = \begin{vmatrix} -p_0 - p_1 - b_0 K_1 - \lambda & p_1 - b_0 K_2 \\ p_2 & -p_2 - \lambda \end{vmatrix}$

$= (-p_0 - p_1 - b_0 K_1 - \lambda)(-p_2 - \lambda) - (p_1 - b_0 K_2)(p_2)$

$= +p_0 p_2 + p_0 \lambda + p_1 p_2 + \lambda p_1 + b_0 K_1 p_2 + b_0 K_1 \lambda + p_2 \lambda + \lambda^2$

$- (\cancel{p_1 p_2} - b_0 K_2 p_2)$

$= \lambda^2 + p_2 \lambda + p_1 \lambda + p_0 \lambda + b_0 K_1 \lambda + p_0 p_2 + b_0 K_1 p_2 + b_0 K_2 p_2$

$= \lambda^2 + (p_2 + p_1 + p_0 + b_0 K_1)\lambda + (p_0 + b_0 K_1 + b_0 K_2) p_2$

eigenvalues $\Rightarrow \lambda_1 = \lambda_2 = -1$

$(\lambda + 1)(\lambda + 1) = \lambda^2 + 2\lambda + 1$

$\lambda^2 + (p_2 + p_1 + p_0 + b_0 K_1)\lambda + (p_0 + b_0 K_1 + b_0 K_2) p_2 = \lambda^2 + 2\lambda + 1$

$p_2 + p_1 + p_0 + b_0 K_1 = 2$

$b_0 K_1 = 2 - p_2 - p_1 - p_0$

$K_1 = \frac{2 - p_2 - p_1 - p_0}{b_0}$

$p_2(p_0 + b_0 K_1 + b_0 K_2) = 1$

$p_2 b_0 K_2 = 1 - p_2(p_0 + b_0 K_1)$

$p_2 b_0 K_2 = 1 - p_2\left(\cancel{p_0} + \frac{\cancel{b_0} + 2 + p_2 + p_1 + \cancel{p_0}}{\cancel{b_0}}\right)$

$K_2 = \frac{1}{p_2 b_0} - \frac{2 + p_2 + p_1}{b_0}$

$\Leftarrow$

$K_2 = \frac{1}{b_0}\left(\frac{1}{p_2} - 2 + p_2 + p_1\right)$

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Q4b) cont.

Control parameter K_c:

$$C(A-BK)^{-1}B = (1\ 0)\begin{pmatrix} 2 & -1 \\ -2-K_1 & 1-K_2 \end{pmatrix}^{-1}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\frac{1}{(2)(1-K_2)-(-1)(-2-K_1)}\begin{pmatrix} 1-K_2 & 1 \\ 2+K_1 & 2 \end{pmatrix}$$

$$= \frac{1}{-2K_2-K_1}\begin{pmatrix} 1-K_2 & 1 \\ 2+K_1 & 2 \end{pmatrix}$$

$$C(A-BK)^{-1}B = (1\ 0)\begin{pmatrix} 1-K_2 & 1 \\ 2+K_1 & 2 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix}\frac{1}{-2K_2-K_1}$$

$$= \frac{1}{-2K_2-K_1}\begin{pmatrix} 1-K_2 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \frac{1}{-2K_2-K_1}(1) = \frac{1}{-2K_2-K_1}$$

$$u = -K_1C_1 - K_2C_2 + 4K_c$$

$$u = -K_1C_1 - K_2C_2 + 4\left(\frac{1}{-2K_2-K_1}\right)$$

where $K_1 = -14$, $K_2 = 6$

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Q4 b) $\frac{dc}{dt} = \begin{pmatrix} 2 & -1 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}u$, $y = (1 \; 0)\, c_1$

$A = \begin{pmatrix} 2 & -1 \\ -2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $C = (1 \; 0)$

$W_r = (B \quad AB) = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$

$W_o = \begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}$

$u = -Kc + K_r r$

$u = -(k_1 \quad k_2)\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + K_r(4)$

$u = -K_1c_1 - K_2c_2 + 4K_r$

Closed loop system becomes

$\frac{dc}{dt} = (A - BK)c + BK_r r$

$BK = \begin{pmatrix} 0 & 0 \\ K_1 & K_2 \end{pmatrix}$

$\frac{dc}{dt} = \begin{pmatrix} 2 & -1 \\ -2-K_1 & 1-K_2 \end{pmatrix} c + 4\begin{pmatrix} 0 \\ K_r \end{pmatrix}$

$y = Cc = c_1$

$\det(A - BK - \lambda I) = \begin{vmatrix} 2-\lambda & -1 \\ -2-K_1 & 1-K_2-\lambda \end{vmatrix}$

$= (2-\lambda)(1-K_2-\lambda) - (-1)(-2-K_1)$

$= \cancel{2} - 2K_2 - 2\lambda - \lambda + \lambda K_2 + \lambda^2 - (\cancel{2} + 2K_1)$

$= \lambda^2 - 3\lambda + \lambda K_2 - 2K_2 - K_1$

$= \lambda^2 + (-3 + K_2)\lambda - 2K_2 - K_1$

Eigenvalues $= \lambda_1 = -2$, $\lambda_2 = -1$

$(\lambda+2)(\lambda+1) = \lambda^2 + 3\lambda + 2$

$\lambda^2 + (-3 + K_2)\lambda - 2K_2 - K_1 = \lambda^2 + 3\lambda + 2$

$-3 + K_2 = 3$ $\qquad$ $-2K_2 - K_1 = 2$

$K_2 = 6$ $\qquad$ $-K_1 = 14$

$K_1 = -14$

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Q3 a) → $R(P) < 1$ it shows the population will decline

→ If $R(P) = 1$ we can find fixed points p^* & P_{crit}

→ $P_{crit} < P < p^*$ means $R(P) > 1$ the population increases but is bounded above by p^*

→ $P > p^*$ then $R(P) < 1$ and the population decreases and is bounded below by p^*.

Q4 a) $\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$, $y = c_1 = (1 \;\; 0)\, c_1$

$A = \begin{pmatrix} 2 & -1 \\ -2 & 1 \end{pmatrix}$ $\quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ $\quad C = (1 \;\; 0)$

Observable

$W_o = \begin{pmatrix} C \\ CA \end{pmatrix}$

$CA = (1 \;\; 0)\begin{pmatrix} 2 & -1 \\ -2 & 1 \end{pmatrix} = (2 \;\; -1)$

$W_o = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}$ $\quad \det(W_o) = -1 \neq 0$

∴ the system is observable as the rows are linearly independent.

Reachable

$W_r = (B \quad AB)$

$AB = \begin{pmatrix} 2 & -1 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$W_r = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$ $\quad \det(W_r) = 1 \neq 0$

∴ the system is reachable